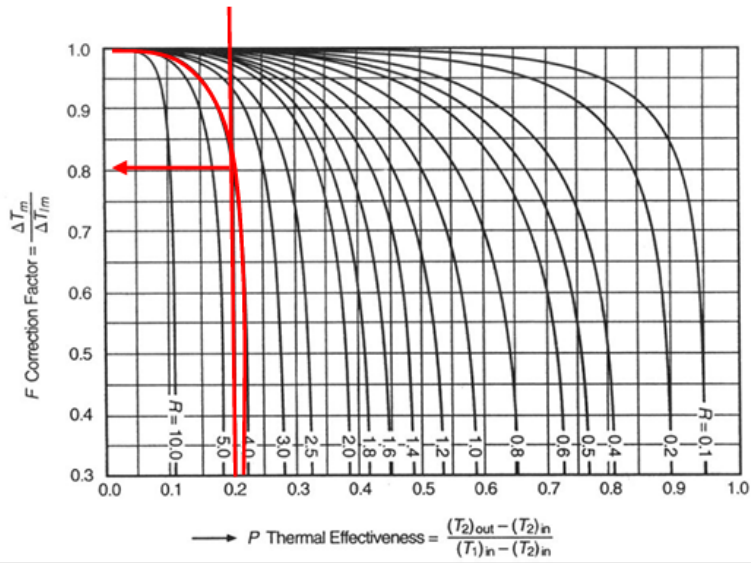
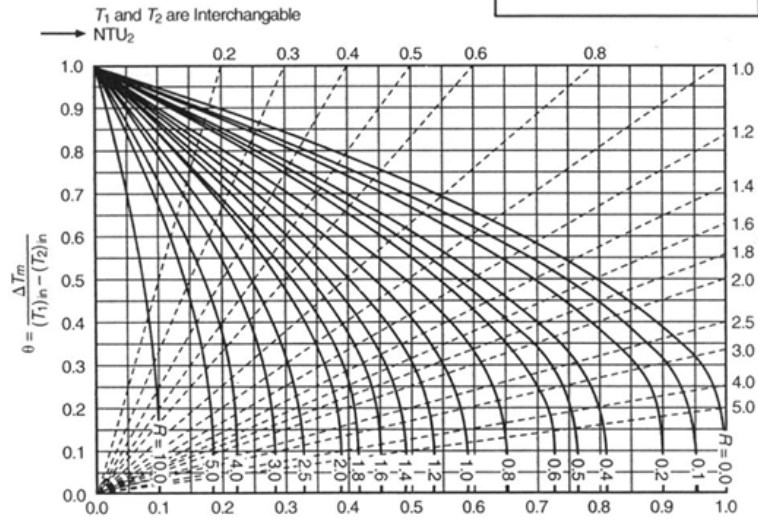
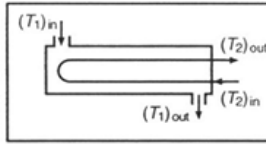


Tema 2 Intercambiadores

Para la selección dependerá de las características operativas que debe cumplir: presiones de trabajo, temperaturas, características de compatibilidad, etc... Una vez seleccionado su cálculo dependerá de los mecanismos de transferencia y de la disposición de los fluido



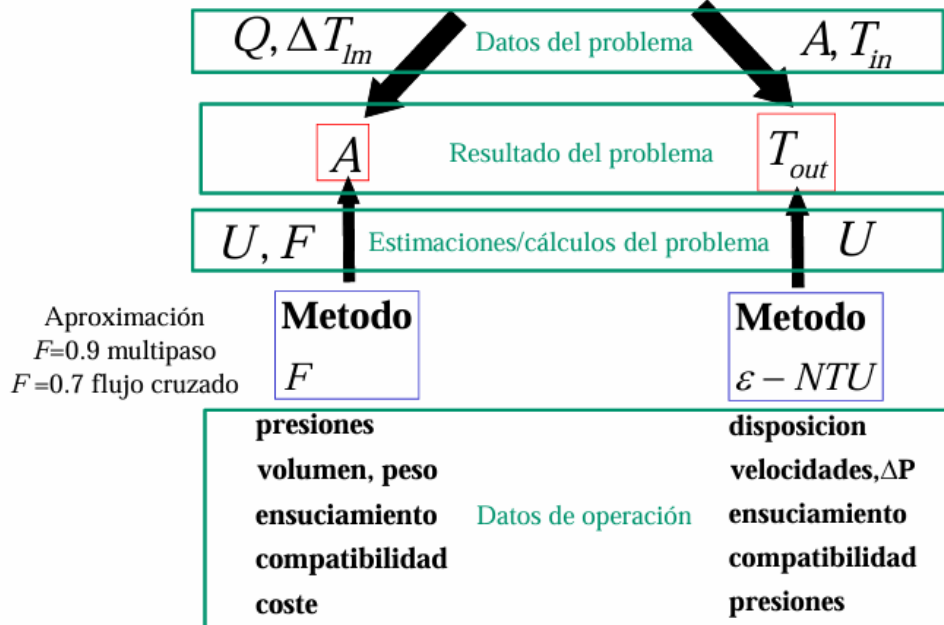


Cálculo y selección

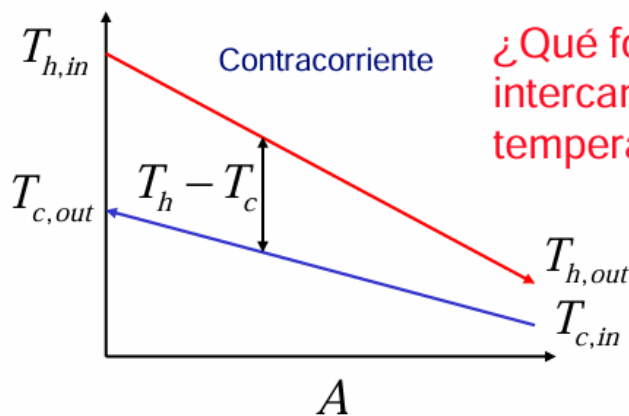
$$Q \Rightarrow mc_p \Delta T_H = mc_p \Delta T_C = mh_{fg}$$

$$F \Delta T_{lm}$$

$$Q = UA \Delta T_m$$



Métodos de cálculo (factor de corrección)



¿Qué fórmula me da el calor total intercambiado en función de las temperaturas de entrada y salida?

$$\frac{dQ}{dA} = -(mc_p)_h \frac{dT_h}{dA}$$

$$\frac{dQ}{dA} = -(mc_p)_c \frac{dT_c}{dA}$$

$$\frac{dQ}{dA} = U(T_h - T_c)$$

$$U(x) \frac{dQ}{dA} = U(x)(T_h(x) - T_c(x))$$

$$\frac{dQ}{dA} \left(\frac{1}{(mc_p)_c} - \frac{1}{(mc_p)_h} \right) = \frac{d(T_h - T_c)}{dA}$$

$$\left(\frac{1}{(mc_p)_c} - \frac{1}{(mc_p)_h} \right) dA = \frac{d(T_h - T_c)}{U(T_h - T_c)}$$

Si el coeficiente no depende de la posición lo que ocurre globalmente también ocurre localmente

$$\int_0^A \left(\frac{1}{(mc_p)_c} - \frac{1}{(mc_p)_h} \right) dA = \int_{T_{h,in}-T_{c,out}}^{T_{h,out}-T_{c,in}} \frac{d(T_h - T_c)}{U(T_h - T_c)} \quad \left(\frac{1}{(mc_p)_c} - \frac{1}{(mc_p)_h} \right) A = \frac{1}{U} \ln \left(\frac{T_{h,out} - T_{c,in}}{T_{h,in} - T_{c,out}} \right)$$

$$Q = (mc_p)_c \Delta T_c = (mc_p)_h \Delta T_h \Rightarrow Q \left(\frac{1}{(mc_p)_c} - \frac{1}{(mc_p)_h} \right) = \Delta T_c - \Delta T_h$$

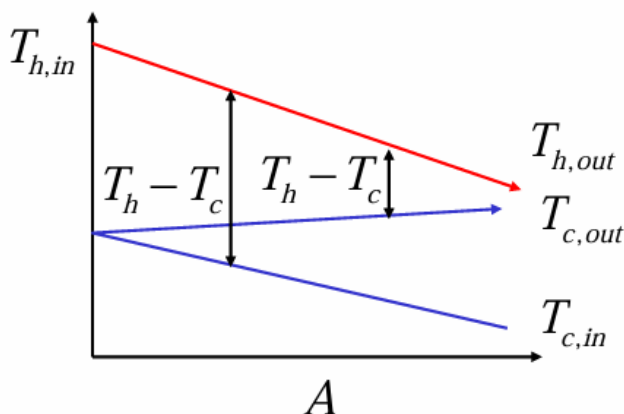
Para un intercambiador en Contracorriente: La media de diferencias de temperaturas es el incremento de temperaturas logarítmico medio

$$Q = UA \frac{\overbrace{\Delta T_h - \Delta T_c}^{\Delta T_m}}{\ln \left(\frac{T_{h,in} - T_{c,out}}{T_{h,out} - T_{c,in}} \right)} = UA \Delta T_{lm}$$

Para cualquier otro intercambiador con diferente disposición de los fluidos su diferencia media de temperaturas se compara con el de contracorriente

contracorriente

$$Q = UA \Delta T_m = UA \Delta T_{lm}$$



Otra disposición

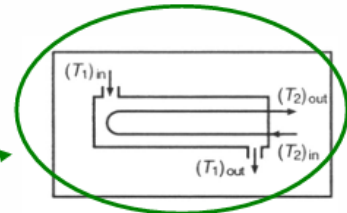
$$Q = UA \Delta T_{m,otra}$$

$$Q = UA \frac{\Delta T_{m,otra}}{\underbrace{\Delta T_{lm}}_F} \Delta T_{lm}$$

$$F \leq 1$$

$$P = \frac{\Delta T_2}{\Delta T_{\max}}$$

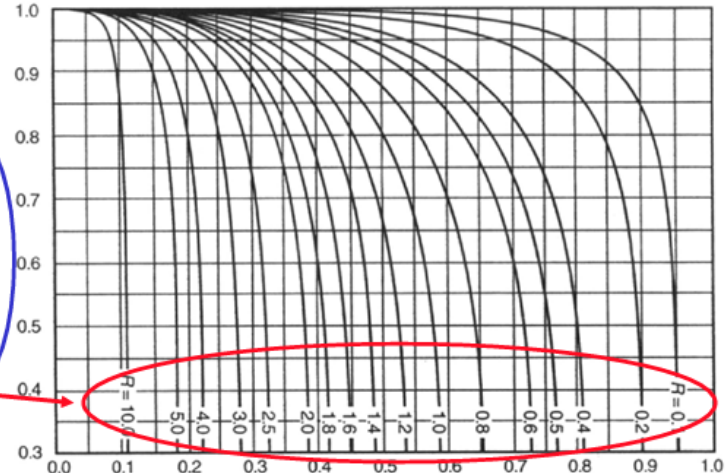
$$R = \frac{C_2}{C_1} = \frac{(mc_p)_2}{(mc_p)_1} = \frac{\Delta T_1}{\Delta T_2}$$



Para cada tipo de intercambiador se grafica frente a la relación de capacidades (R) y a la eficiencia térmica (P)

F

$$F \text{ Correction Factor} = \frac{\Delta T_m}{\Delta T_{lm}}$$

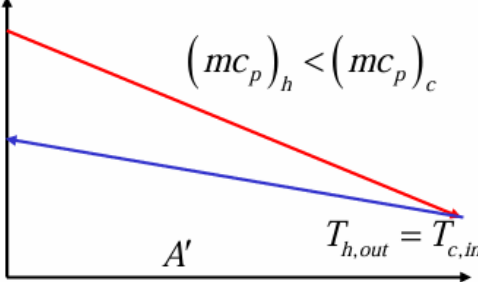
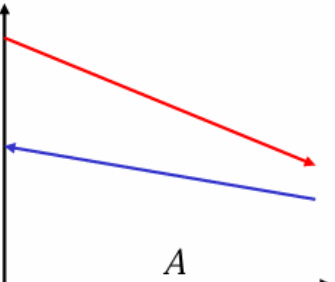


$$P \text{ Thermal Effectiveness} = \frac{(T_2)_{out} - (T_2)_{in}}{(T_1)_{in} - (T_2)_{in}}$$

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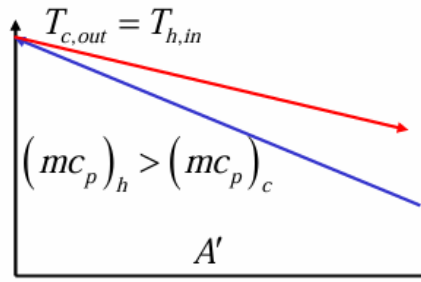
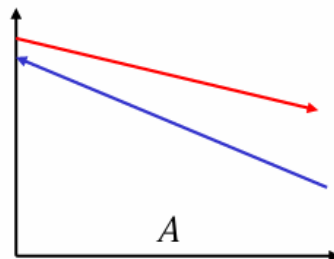
Eficiencia de un intercambiador (ε)

$$\varepsilon = \frac{Q}{Q_{\max}}$$



$$Q_{\max} = (mc_p)_h (T_{h,in} - T_{h,out}) = (mc_p)_h (T_{h,in} - T_{c,in})$$

$$Q_{\max} = (mc_p)_{\min} (T_{h,in} - T_{c,in})$$



$$Q_{\max} = (mc_p)_c (T_{c,out} - T_{c,in}) = (mc_p)_c (T_{h,in} - T_{c,in})$$

$$\varepsilon = P \text{ si } (mc_p)_2 = (mc_p)_{\min} ; \varepsilon = PR \text{ si } (mc_p)_1 = (mc_p)_{\min}$$

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$$\left(\frac{1}{(mc_p)_2} - \frac{1}{(mc_p)_1} \right) AU = \ln \left(\frac{T_{1,out} - T_{2,in}}{T_{1,in} - T_{2,out}} \right)$$

$$\left(1 - \frac{(mc_p)_2}{(mc_p)_1} \right) \frac{NTU_2}{(mc_p)_2} = \ln \left(\frac{T_{1,out} - T_{2,in}}{T_{1,in} - T_{2,out}} \right)$$

$$\frac{T_{2,out} - T_{2,in}}{T_{1,in} - T_{2,in}} = P, \quad \frac{T_{1,in} - T_{2,out}}{T_{1,in} - T_{2,in}} = 1 - P, \quad \frac{T_{1,out} - T_{2,in}}{T_{1,in} - T_{2,in}} = 1 - RP$$

$$\exp[(1 - R) NTU] = \frac{1 - RP}{1 - P}$$

$$f(R, NTU, P) = 0 \text{ otras configuraciones}$$

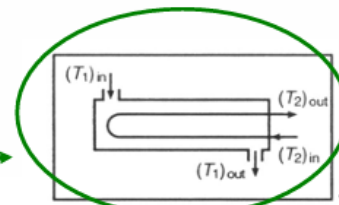
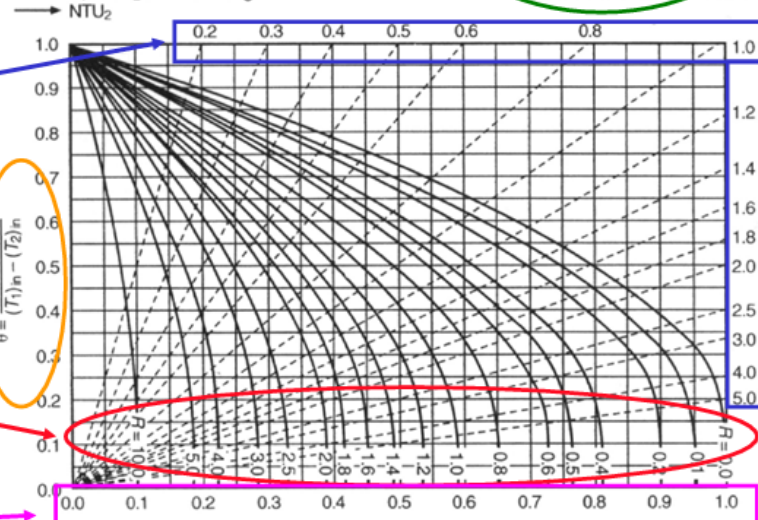
$$\theta = \frac{F \Delta T_{lm}}{\Delta T_{max}}$$

$$NTU_2 = \frac{UA}{C_2}$$

Para cada tipo de intercambiador se grafica NTU y θ

frente a la relación de capacidades (R) y a la eficiencia térmica (P)

T_1 and T_2 are interchangeable



Depende del tipo de fluidos en el intercambiador

Aproximación $F=0.9$ multitubulares, $F=0.7$ flujo cruzado

$$C = \exp \left(\ln C_1 + \ln \left(C_1 / C_2 \right) \frac{\ln \left[(Q / \Delta T_m) / (Q / \Delta T_m)_1 \right]}{\ln \left[(Q / \Delta T_m)_1 / (Q / \Delta T_m)_2 \right]} \right)$$

